INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

ME 361
DYNAMICS OF MACHINERY

Lecture 1

INTRODUCTION, REVIEW OF TRIG. AND VECTORS, AND UNIT 1

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Mechanical Engineering Department, KNUST

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Grading:

- Continuous Assessment 30%
- Final Exam 70%

Prerequisite: A previous course in ME 161/162/ and ME 260.

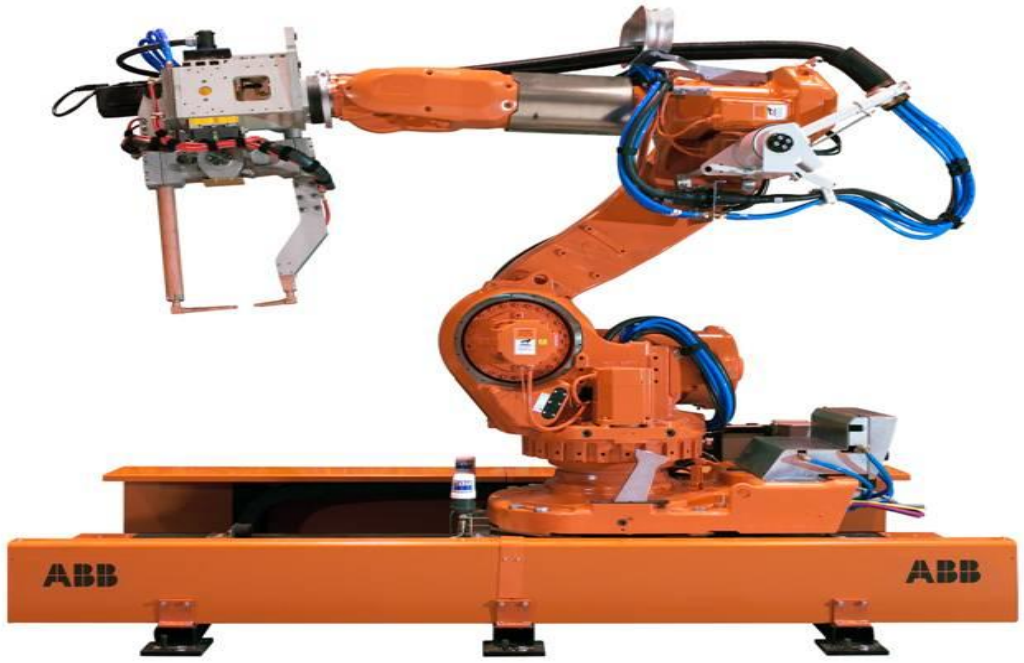
Homework: About 3-4 problems will be assigned every other week and these will be collected for grading. Collaboration is encouraged, but independent work is required and essential in order to perform well on the exams.

REFERENCE BOOKS

- Mechanics of Machines, Advanced Theory and Examples, John Hannah, R. C. Stephens.
- Mechanics of Machines, Elementary Theory and Examples, John Hannah, R. C. Stephens.
- Theory of Machines, R.S. Khurmi and J.K. Gupta, 2004
- Kinematics and Dynamics of Machinery, Charles E. Wilson and J. Peter Sadler.
- VECTOR MECHANICS FOR ENGINEERS (Statics and Dynamics), Ferdinand P. Beer, E. Russell Johnston, Jr., David F. Mazurek, Phillip J. Cornwell, Elliot R. Eisenberg

COURSE INTRODUCTION

- Many solutions to engineering problems require the application of the principles of dynamics e.g. Design of vehicular structural (cars, aircraft, ships), steering system, suspension, etc.
- This is also true for many mechanical devices such as automatic switches, motors, pumps, turbines, vibrators and industrial machines.
- In addition, common household items such switches, watches, calculators, blenders, microwaves require dynamic analysis.
- investigations of vehicular accidents and of predictions of the motions of artificial satellites, projectiles, and space craft are based on principles of dynamics.


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CARNIVAL MAGIC

COURSE OUTLINE

- Review of Trigonometry and Vectors
- Unit 1: Rigid bodies motions and applications
- Unit 2: Geometric properties of rigid bodies
- Unit 3 :Gyroscopic motion and its effect effects in machinery
- Unit 4: Static and inertia forces and torques in mechanisms.
- Unit 5: Fluctuation of energy and speed in machines, crank-effort and turning moment diagrams, flywheels and governors
- Unit 6: Balancing of multi-cylinder engines including radial and V-engines.

REVIEW OF TRIGONOMETRY

- You must have mastered right-triangle trigonometry

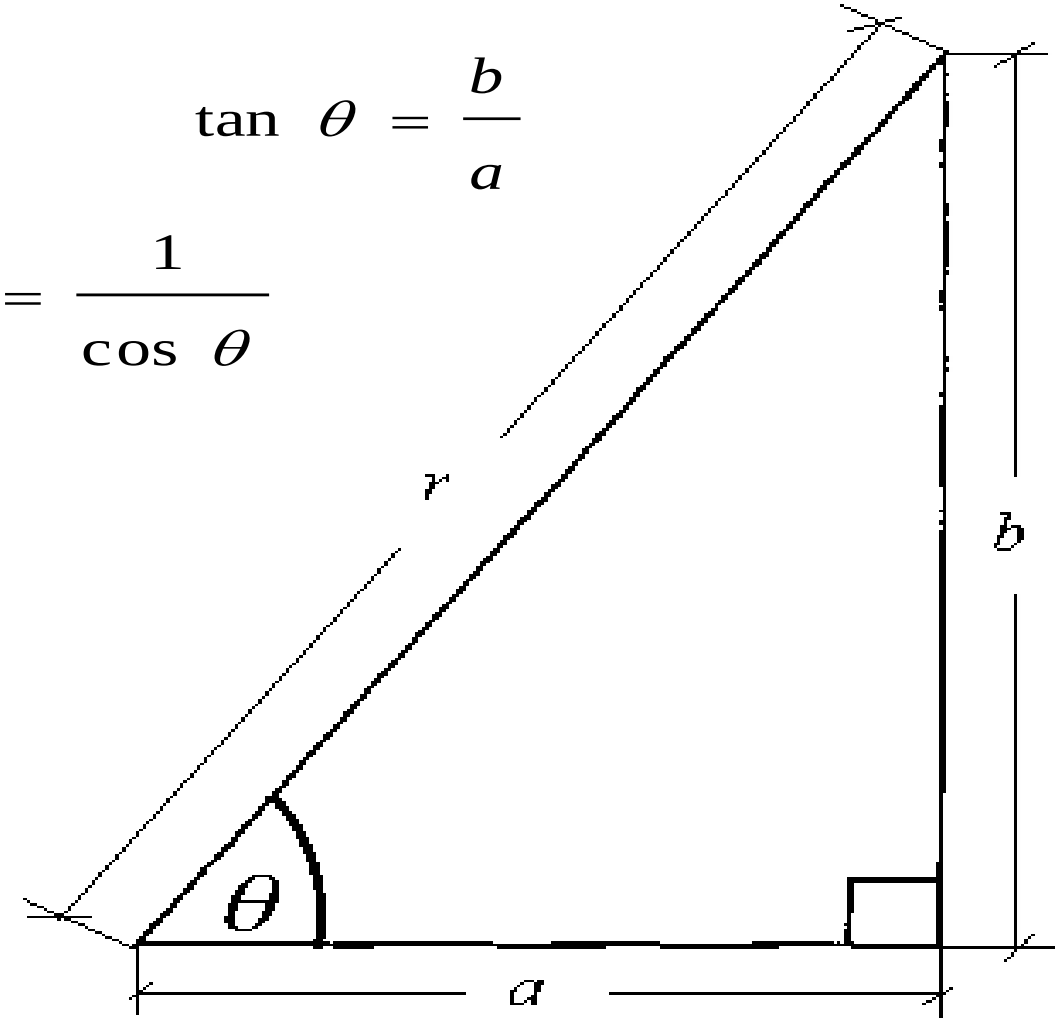
$$\sin\theta = \frac{b}{r} \qquad \cos\theta = \frac{a}{r} \qquad \tan\theta = \frac{b}{a}$$

$$\text{cosec}\ \theta = \frac{1}{\sin\theta} \qquad \text{secant}\ \theta = \frac{1}{\cos\theta}$$

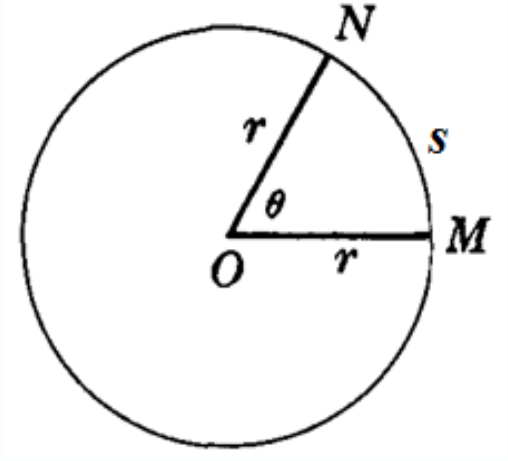
$$\text{cotan}\ \theta = \frac{1}{\tan\theta}$$

Pythagoras Theorem

$$r^2 = a^2 + b^2$$



RELATIONSHIP BETWEEN DEGREES AND RADIANS



The arc s described when the line ON rotates through 360° is $2\pi r$

i.e. $2\pi r \equiv 360^o \quad \therefore s \equiv \theta^\circ \Rightarrow \quad s = \frac{\theta^\circ}{360} \times 2\pi r$

$$s = r\theta$$

$$\theta \text{ rad} = \frac{\theta^\circ}{180}\pi$$

- 1 radian = 180°/ π = 57.29577 95130 823°
- 1° = π /180 radians = 0.01745 32925 radians

RELATIONSHIPS AMONG TRIGONOMETRIC FUNCTIONS

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

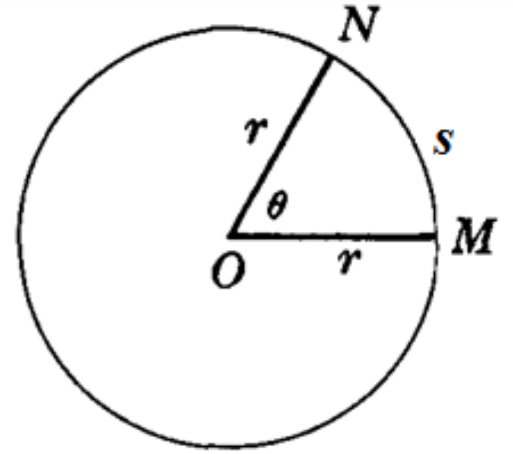
$$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin B \sin A$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan B \tan A}$$

$$\cot(A \pm B) = \frac{\cot A \cot B \mp 1}{\cot B \mp \cot A}$$

SMALL ANGLES



If θ is small;

$\sin\theta \approx \tan\theta \approx \theta$

and $\cos\theta \approx 1$

SINE AND COSINE RULES

SINE RULE

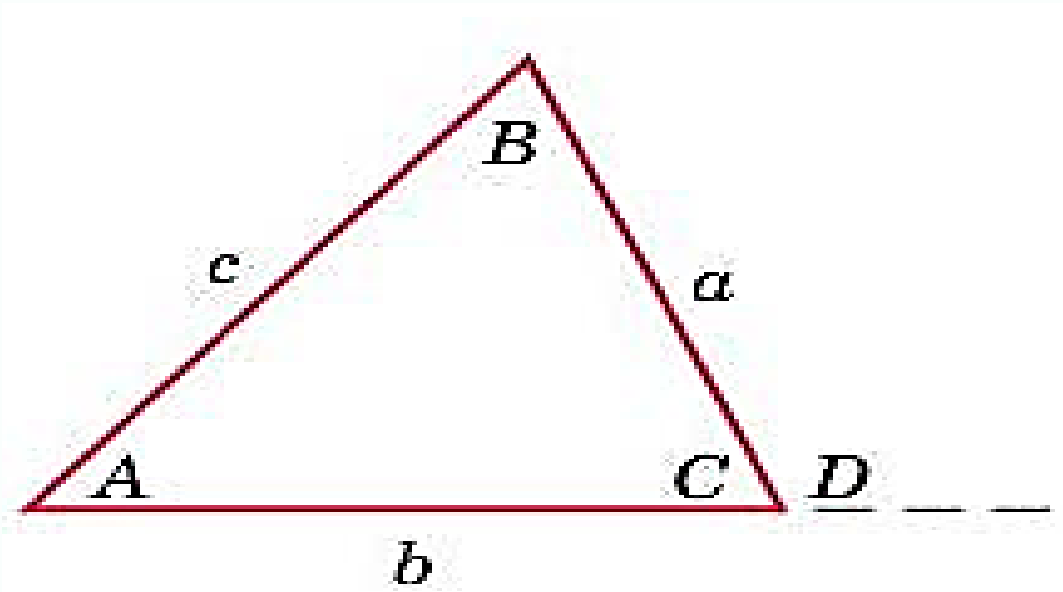
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

COSINE RULE

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$



REVIEW OF VECTORS

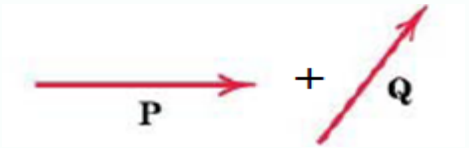
- A vector is a quantity that has both direction and magnitude.

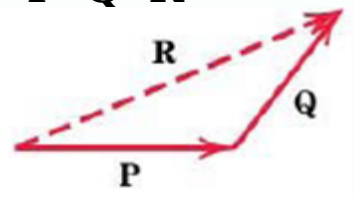
NOTATION

Vector quantities are printed in boldface type, and scalar quantities appear in lightface italic type. Thus , the vector quantity **V** has a scalar V. In long hand work vector quantities should always be consistently indicated by a symbol such as $\underline{V}$ or $\vec{V}$ to distinguish them from scalar quantities.

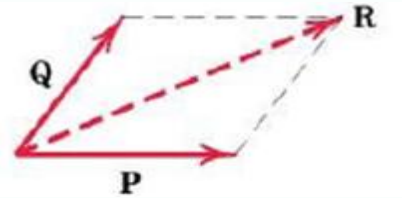
Addition

P+Q=R





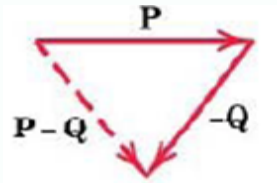
Parallelogram addition



Subtraction



P-Q=P+(-Q)



Commutative law **P+Q=Q+P**

Associative law **P+(Q+R)=(P+R)+Q**

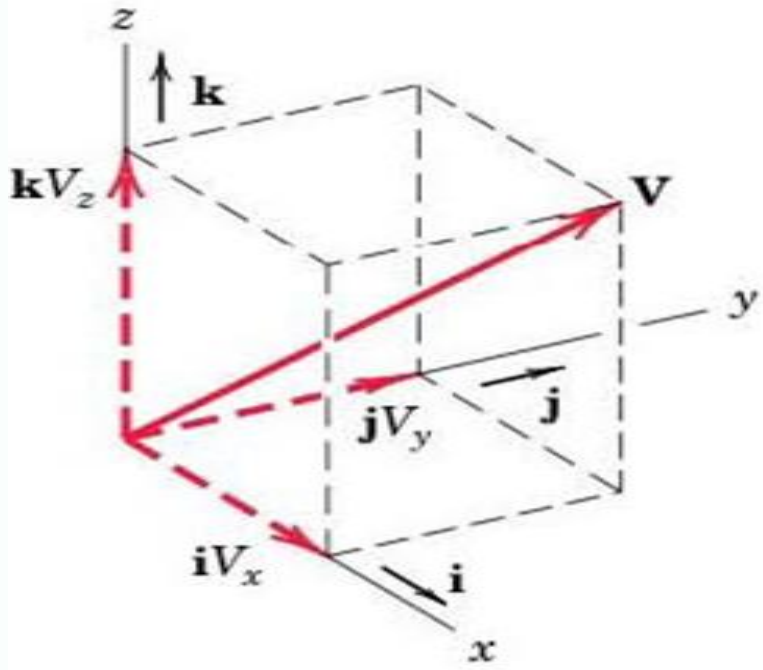
VECTOR DECOMPOSITION

Unit vectors ***i, j, k***

$|i|=|j|=|k|=1$

$$V = V_x\boldsymbol{i} + V_y\boldsymbol{j} + V_z\boldsymbol{k}$$

$$|\boldsymbol{V}| = V = \sqrt{V_x{}^2 + V_y{}^2 + V_z{}^2}$$



direction cosines *l, m, n are the direction cosines of the angles between V and the x, y, z-axes. Thus,*

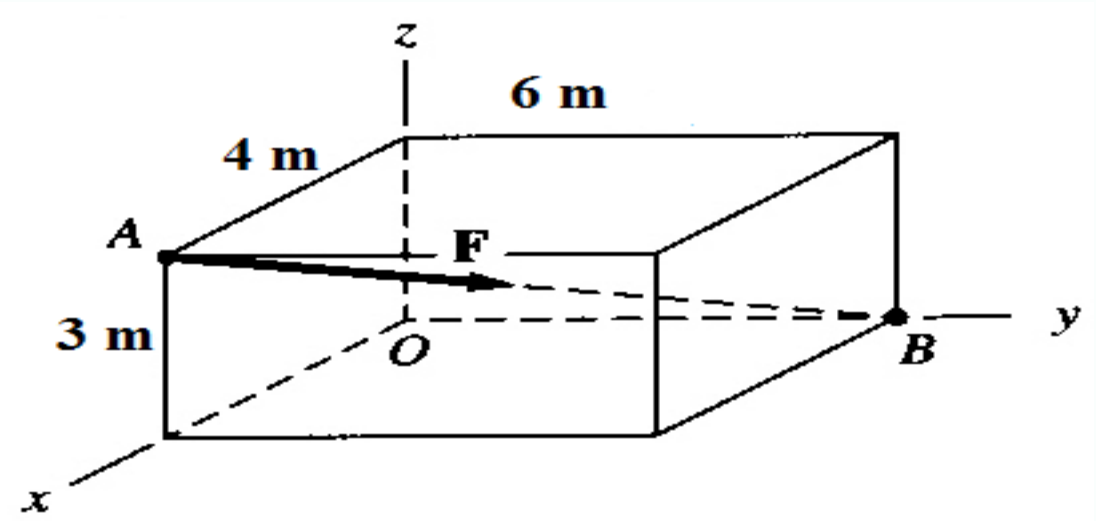
$l=V_x/V$ $\quad m=V_y/V$ $\quad n=V_z/V$

So that $\boldsymbol{V}=V(l\boldsymbol{i}+m\boldsymbol{j}+n\boldsymbol{k})$

$$l^2 + m^2 + n^2 = 1$$

EXAMPLE

Determine the rectangular representation of the 200 N force, F,



$$F = |F| \frac{-4\boldsymbol{i} + 6\boldsymbol{j} - 3\boldsymbol{k}}{\sqrt{4^2 + 6^2 + 3^2}}$$

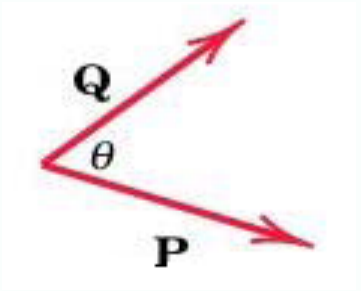
$$F = 200 \frac{-4\boldsymbol{i} + 6\boldsymbol{j} - 3\boldsymbol{k}}{\sqrt{4^2 + 6^2 + 3^2}}$$

$$F = 200 \frac{-4\boldsymbol{i} + 6\boldsymbol{j} - 3\boldsymbol{k}}{\sqrt{\boldsymbol{61}}}$$

$$F = -102.43\boldsymbol{i} + 153.65\,\boldsymbol{j} - 76.83\boldsymbol{k} \quad \boldsymbol{N}$$

DOT/SCALAR PRODUCTS

$$\boldsymbol{P.Q} = |P||Q|cos\theta$$



$from\ the\ definition\ of\ the\ dot\ product$

$$\boldsymbol{i.i} = |i||i|cos0 = 1 = \boldsymbol{j.j} = \boldsymbol{k.k}$$

Also $\boldsymbol{i.j} = |i||j|cos90 = 0 = \boldsymbol{j.i} = \boldsymbol{j.k}$

$$P.Q = (P_x\boldsymbol{i} + P_y\boldsymbol{j} + P_z\boldsymbol{k}).(Q_x\boldsymbol{i} + Q_y\boldsymbol{j} + Q_z\boldsymbol{k}) = P_xQ_x + P_yQ_y + P_zQ_z$$

Example

A=8**i**-3**j**-5**k** and B=4**i**-6**j**+5**k**

A.B=(8**i**-3**j**-5**k**).(4**i**-6**j**+5**k**)

=32+18-25

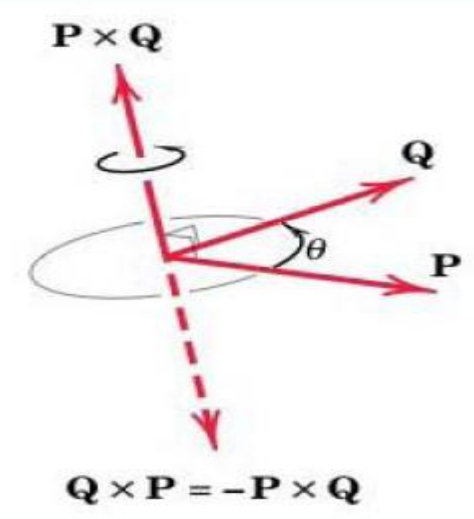
= 25

CROSS/VECTOR PRODUCTS

$$\boldsymbol{P} \times \boldsymbol{Q} = |P||Q|sin\theta\, \boldsymbol{n}$$

$$\boldsymbol{Q} \times \boldsymbol{P} = -\boldsymbol{P} \times \boldsymbol{Q}$$

Distributive law $\quad \boldsymbol{P} \times (\boldsymbol{Q} + \boldsymbol{R}) = \boldsymbol{P} \times \mathbf{Q} + \mathbf{P} \times \mathbf{R}$

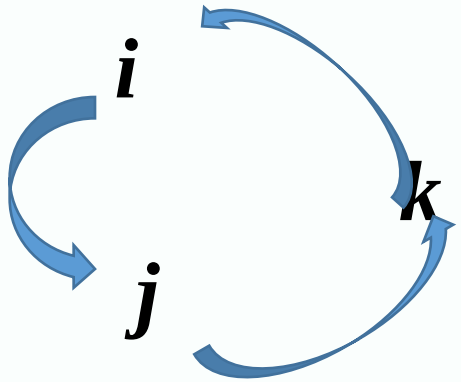


From the definition of the cross product, using a righ − handed coordinate system, we get

$$\boldsymbol{i} \times \boldsymbol{j} = |\boldsymbol{i}||\boldsymbol{j}|\boldsymbol{sin90}\, \boldsymbol{k} = \boldsymbol{k}$$
$$\boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i}; \quad \boldsymbol{k} \times \boldsymbol{i} = \boldsymbol{j}$$

Also $\quad \boldsymbol{i} \times \boldsymbol{k} = -\boldsymbol{j}; \quad \boldsymbol{k} \times \boldsymbol{j} = -\boldsymbol{i}; \boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}$

and $\quad \boldsymbol{i} \times \boldsymbol{i} = \boldsymbol{j} \times \boldsymbol{j} = \boldsymbol{k} \times \boldsymbol{k} = \mathbf{0}$



CROSS/VECTOR PRODUCTS cont'd…

$$P \times Q = (P_x i + P_y j + P_z k) \times (Q_x i + Q_y j + Q_z k)$$

$$P \times Q = (P_y Q_z - P_y Q_z) i + (P_z Q_x - P_x Q_z) j + (P_x Q_y - P_y Q_x) k$$

The cross product may also be represented by the determinant of

$$\begin{vmatrix} i & j & k \\ P_x & P_y & P_z \\ Q_x & Q_x & Q_x \end{vmatrix}$$

CROSS/VECTOR PRODUCTS cont'd…

Example; $A = 8i - 3j - 5k$ *and* $B = 4i - 6j + 5k$

$$A \times B = (8i - 3j - 5k) \times (4i - 6j + 5k)$$

$$= 32(i \times i) - 48(i \times j) + 40(i \times k) - 12(j \times i) + 18(j \times j)$$

$$- 15(j \times k) - 20(k \times i) + 30(k \times j) - 25(k \times k)$$

$$= 0 - 48k - 40j + 12k + 0 - 15i - 20j - 30i - 0 = -45i - 60j - 36k$$

Alternativ ely

$$A \times B = \begin{vmatrix} i & j & k \\ 8 & -3 & -5 \\ 4 & -6 & 5 \end{vmatrix} = i\begin{vmatrix} -3 & -5 \\ -6 & 5 \end{vmatrix} - j\begin{vmatrix} 8 & -5 \\ 4 & 5 \end{vmatrix} + k\begin{vmatrix} 8 & -3 \\ 4 & -6 \end{vmatrix}$$

$$= -45\,i - 60\,j - 36\,k$$

UNIT 1: THREE-DIMENSIONAL KINEMATICS OF RIGID BODIES

- **Session 2-1: Types of Rigid Body Motions**
  - 1-1.1 Pure Translation
  - 1-1.2 Pure Rotation
  - 1-1.3 General Motion
- **Session 2-2: General Rotational Motion**
  - 1-2.1 Euler's Theorem of Rotation
  - 1-2.2 Finite Rotations
  - 1-2.3 Infinitesimal Rotation
  - 1-2.4 Angular Velocity
  - 1-2.5 Linear Velocity
  - 1-2.6 Angular Acceleration
  - 1-2.7 Linear Acceleration
  - 1-2.8 Transport Theorem
- **Session 2-3: General Motion**
  - 1-3.1 Position and Displacement
  - 1-3.2 Velocity
  - 1-3.3 Acceleration

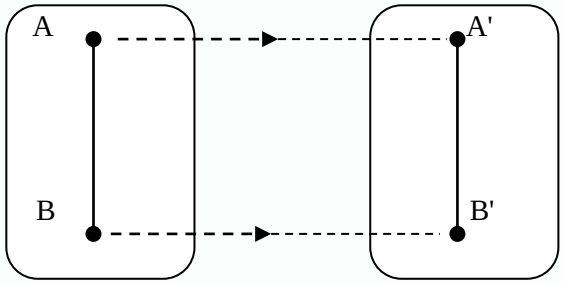
Types of Rigid Body Motion

Motions of a rigid body may be classify into three categories:

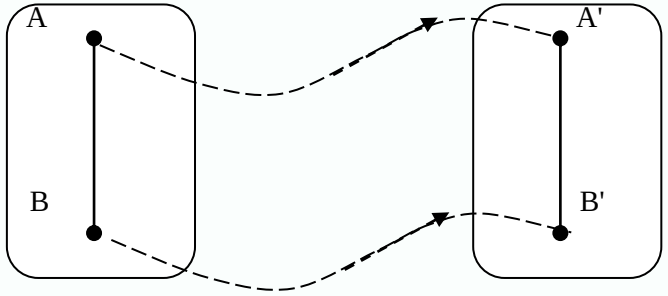
- Pure Translation
- Pure Rotation
- General Motion

Pure Translation

- In a pure translational motion a body moves such that the body remains parallel to its past and future positions at all times.
- In rectilinear translation, all paths of points on the body are parallel and in straight lines .
- A rigid body may translate along a curved path without any rotation. E.g. motion exhibited by an aircraft during landing



Rectilinear Translational Motions



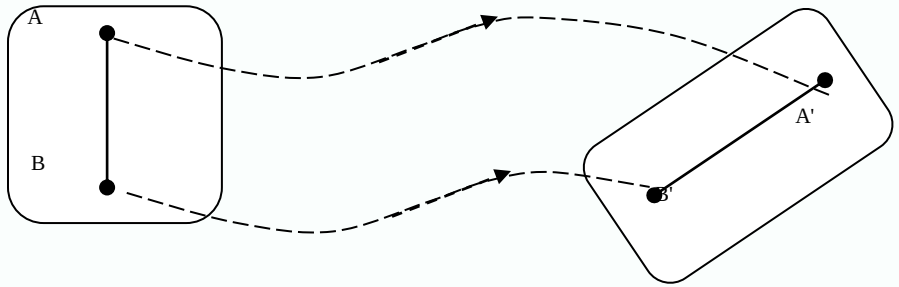
Curvilinear Translational Motions

Pure Rotation

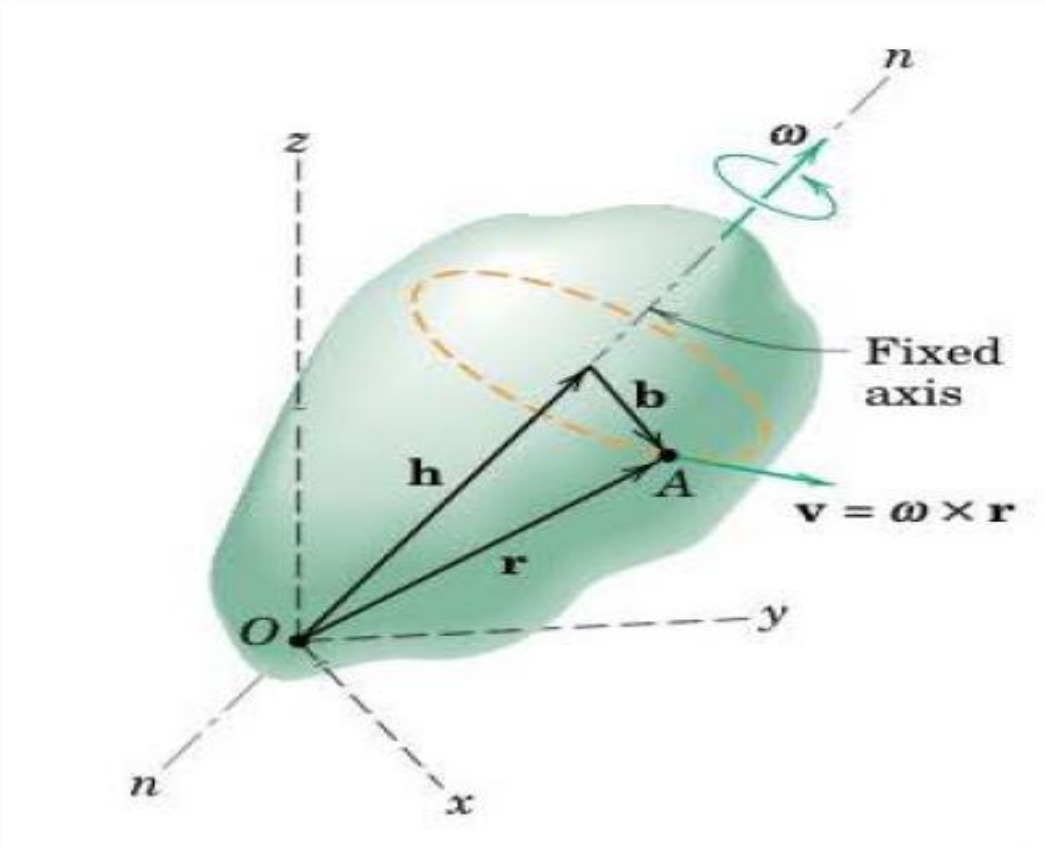
- It is the angular motion about a fixed point or line. Rotational motion is described using only rotational parameters such angular displacement, angular velocity and angular acceleration.
- Types of Pure Rotation
  - Rotation about a fixed axis
  - Rotation about a fixed point

General Motion

- General motion is involves combination of translational and rotational motions.
- i.e. General (or complex) motion = Translation motion+ Rotational Motion.
- An unrestricted three-dimensional motion of a rigid body has six degrees of freedom, which are translation and rotation about x, y and z axis.



Fixed axis rotation



- The rotation axis in general is a fixed axis in the body.
- The angular distance, angular velocity and angular accelerations do not change their directions since they lie along a specified axis.

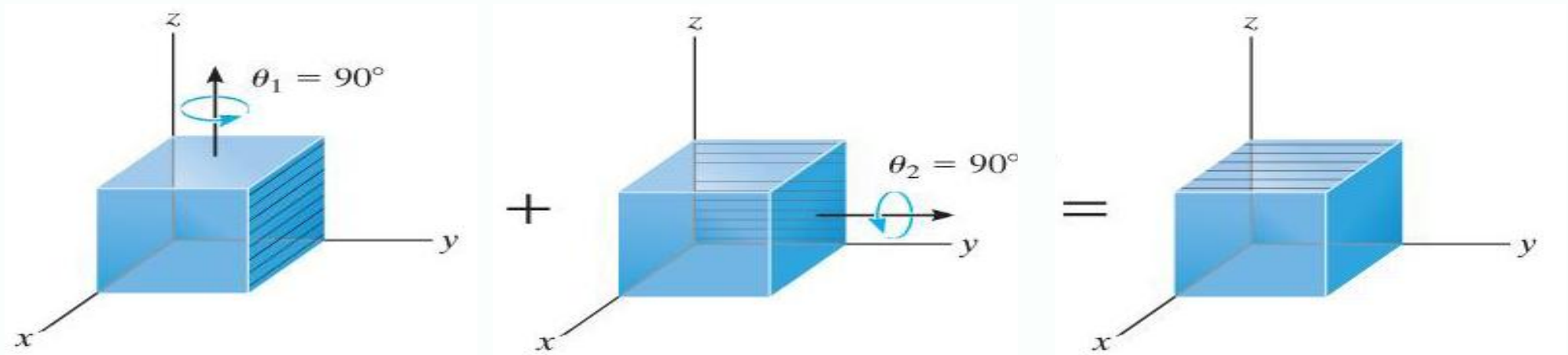
$$s = r\theta$$

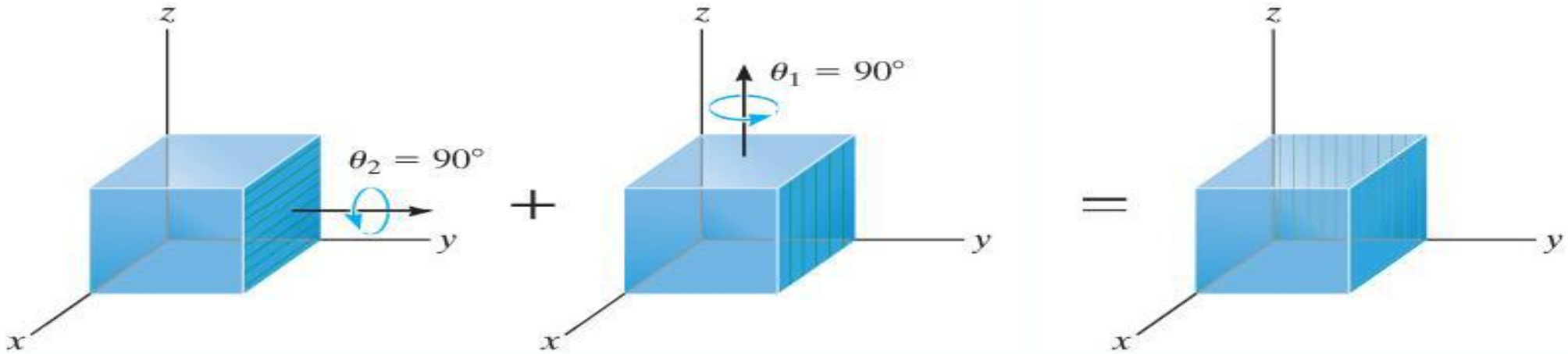
$$v = r\omega$$

$$a = r\alpha$$

Rotation about fixed point(Finite Rotations)

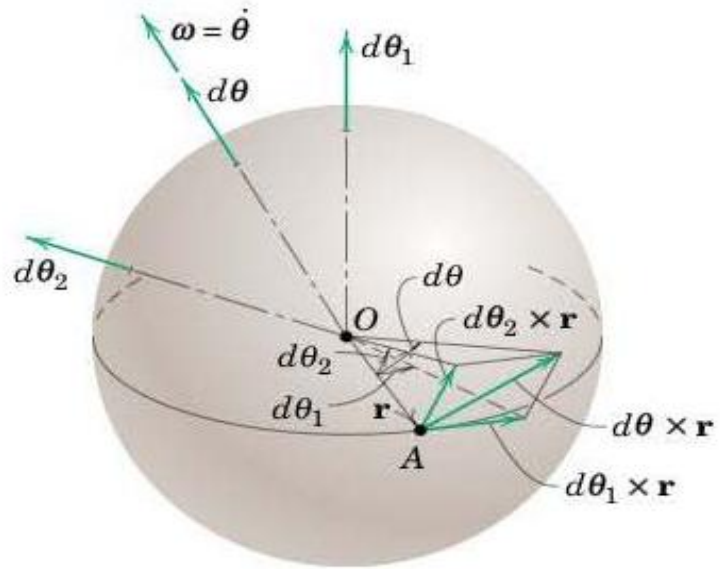
- **Euler's theorem states that two components of rotations about a different axes passing through a point are equivalent to a single rotation about an axis passing through the point.**
- **Finite rotates are not classified as vectors and they do not obey Euler's law of rotation. Also, finite rotations are not commutative.**





Rotation about a fixed point (Infinitesimal Rotations)

In defining angular motions of a body subjected to 3D motion, only small rotations are considered. Such rotations may be classified as vectors and they obey Euler's law of rotation. The resultant rotation and angular velocity of two infinitesimal rotations of line OA is given as



$$d\theta = d\theta_1 + d\theta_2$$

$$\omega = \omega_1 + \omega_2$$

$$\alpha = \alpha_1 + \alpha_2$$

where

$$\omega = \frac{d\theta}{dt} = \dot{\theta}$$

$$\alpha = \dot{\omega} = \dot{\theta}$$

At any instant of time a body with only one fixed point is rotating instantaneously about a particular axis passing through this fixed point.

Transport Theorem

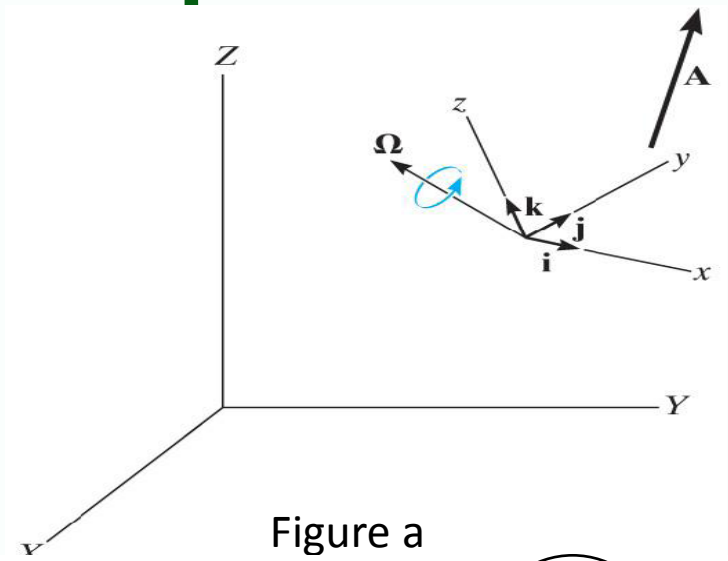


Figure a

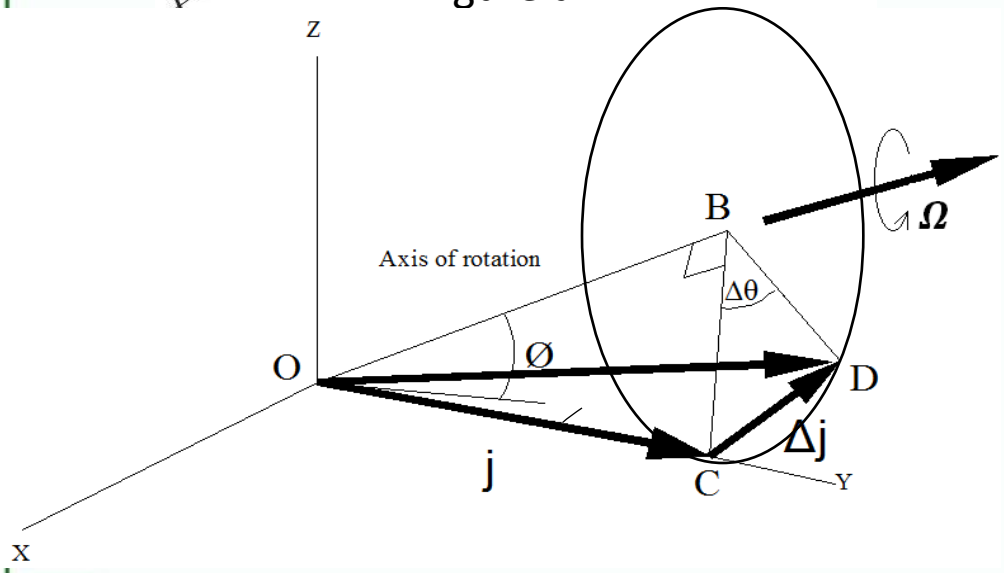


Figure b

- Consider a vector **A** observed from a moving coordinate system *xyz*, which is rotating with angular velocity Ω

$$\vec{A} = A_x i + A_y j + A_k k \qquad \dot{\vec{A}} = \dot{A}_x i + \dot{A}_y j + \dot{A}_k k + A_x \dot{i} + A_y \dot{j} + A_k \dot{k}$$

Therefore

$$If \quad \dot{A}_{rel} = \dot{A}_x i + \dot{A}_y j + \dot{A}_k k \qquad \dot{\vec{A}} = \dot{A}_{rel} + \Omega \times \vec{A}$$

$$\dot{\vec{A}} = \dot{A}_{rel} + A_x \dot{i} + A_y \dot{j} + A_k \dot{k}$$

From Figure b,

$$\Delta j = BC\,\Delta\theta = j \sin\phi\,\Delta\theta$$

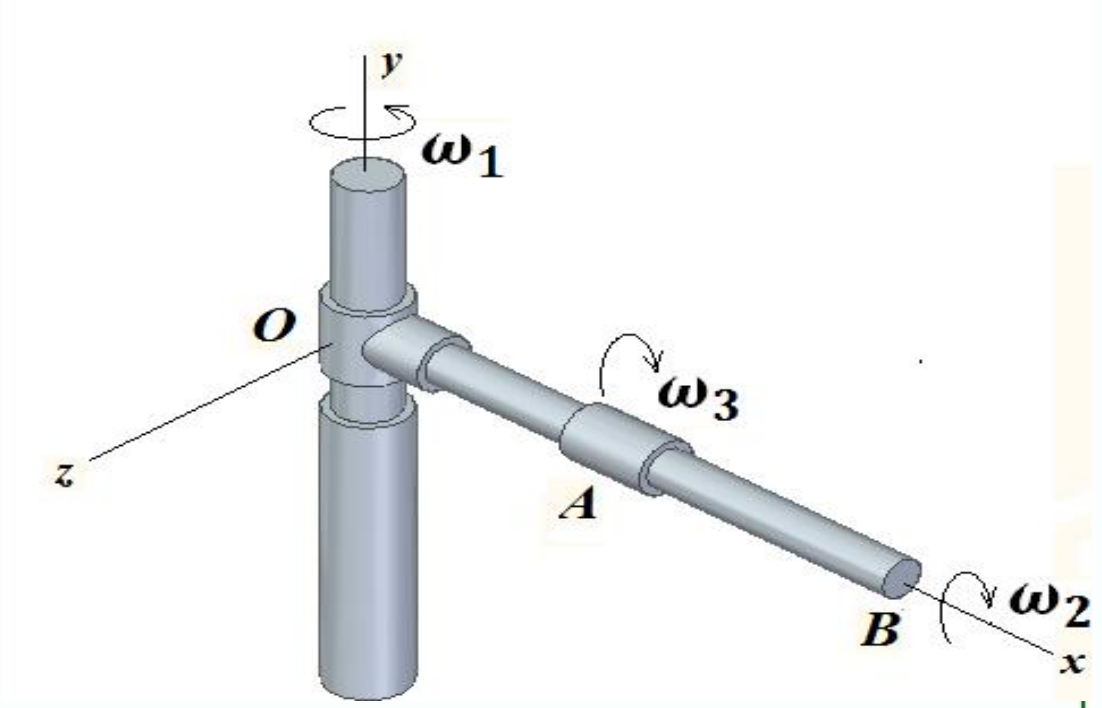
$$\lim_{\Delta t \to 0} \frac{\Delta j}{\Delta t} = \dot{j} = \lim_{\Delta t \to 0} \left| \frac{\Delta j}{\Delta t} \right| |j| \sin\phi = \Omega \times j$$

Generally ,

$$\dot{i} = \Omega \times j, \quad \dot{j} = \Omega \times j \quad and \quad \dot{k} = \Omega \times k$$

Example 1-1

Find the angular velocity and angular acceleration of sleeve A.



Solution

Note that the shaft is not carried by any part, whereas the arm is carried by the shaft and the sleeve is carried by both the shaft and the arm. The relative angular accelerations are determined separately, and then added. Using the transport theorem Equation , the angular velocity of the shaft is

$$\omega = \omega_1 + \omega_2 + \omega_3$$

similarly , the angul ar acceler ation is

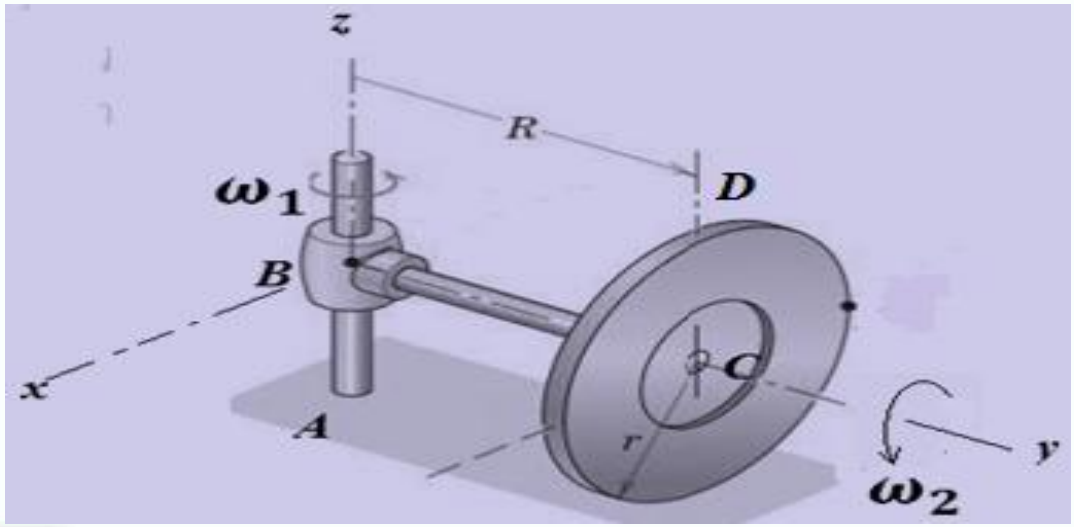
$$\alpha = \alpha_1 + \alpha_2 + \alpha_3$$

$$\alpha = (\dot{\omega}_1) + (\dot{\omega}_2 + \omega_1 \times \omega_2) + (\dot{\omega}_3 + \omega_1 \times \omega_3 + \omega_2 \times \omega_3)$$

Example 1-2

The horizontal shaft BC is rigidly attached to the vertical shaft AB. The vertical shaft AB rotates at a constant angular velocity ω_1 = 4 rad/s about the vertical axis while the disc mounted at the end of BC is rotates about the horizontal axis at ω_2 = 1 rad/s and α_2 =2 rad/s^2. Determine the angular velocity and acceleration of the disc.

Given that R = 1.5 m, r = 0.5 m



Solution

The angular velocity of point D is simply the vector addition of angular velocities of shaft BC about AB and disc about point C. Thus,

$$\omega = \omega_1 + \omega_2 = (4k) + (1j) \quad \omega = j + 4k \text{ (rad/s)}$$

Shaft BC is transporting the disc, while the disc and the shaft are rotating independently. Using the transport theorem,

$$\alpha = (\dot{\omega}_1 + \dot{\omega}_2)_{x,y,z} + \omega_1 \text{ x } \omega_2 = (0 + 2j) + 4k \text{ x } j$$

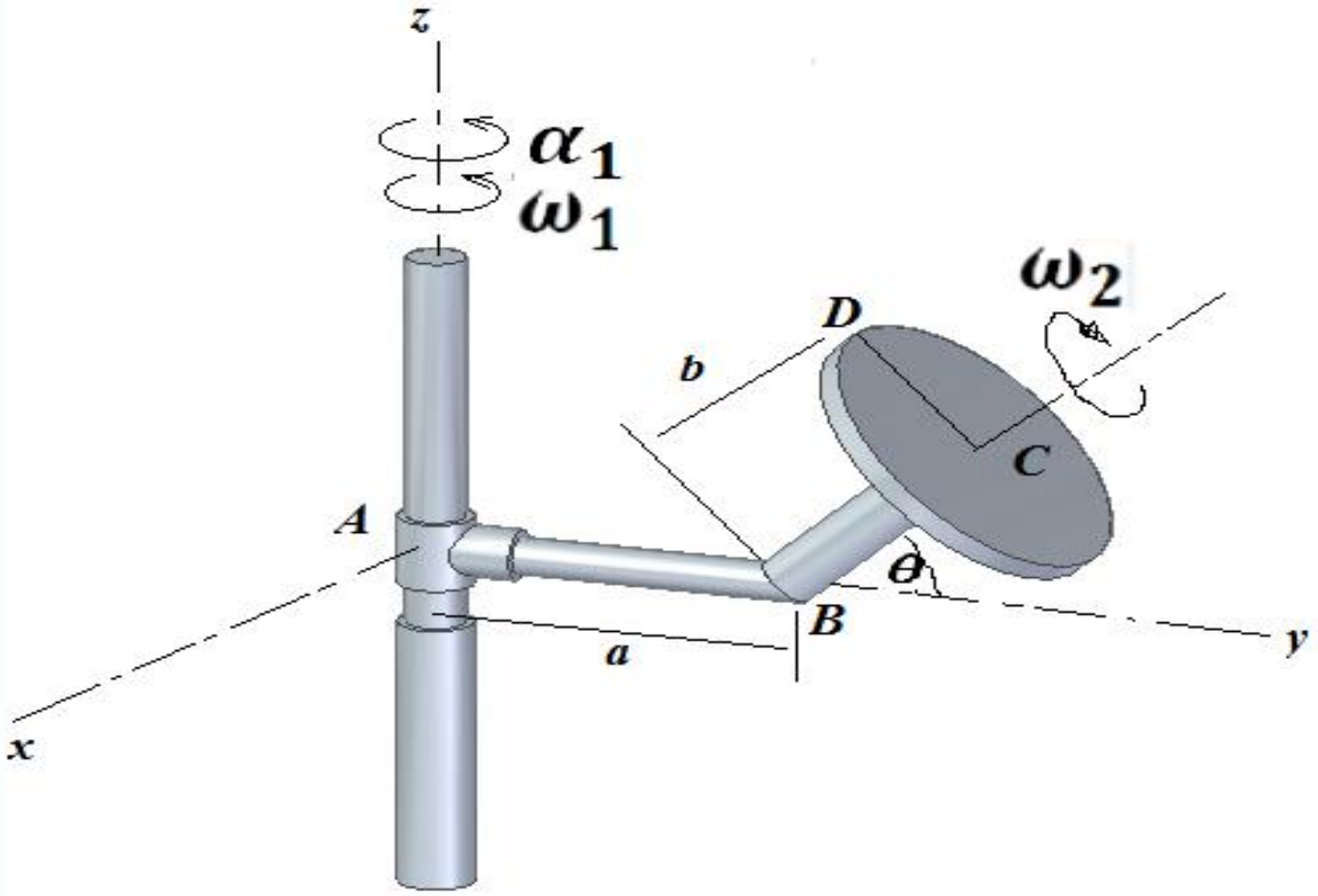
$$\alpha = -4i + 2j \quad (\text{rad/s}^2)$$

Example 1-3

The bent pipe ABC is rigidly attached to a vertical shaft, which rotates with angular velocity ω_1 = 3 rad/s and angular acceleration α_1 = 1 rad/s^2 relative to the ground. The circular disc mounted at point C rotates with constant angular velocity ω_2 = 5 rad/s relative to the bar.

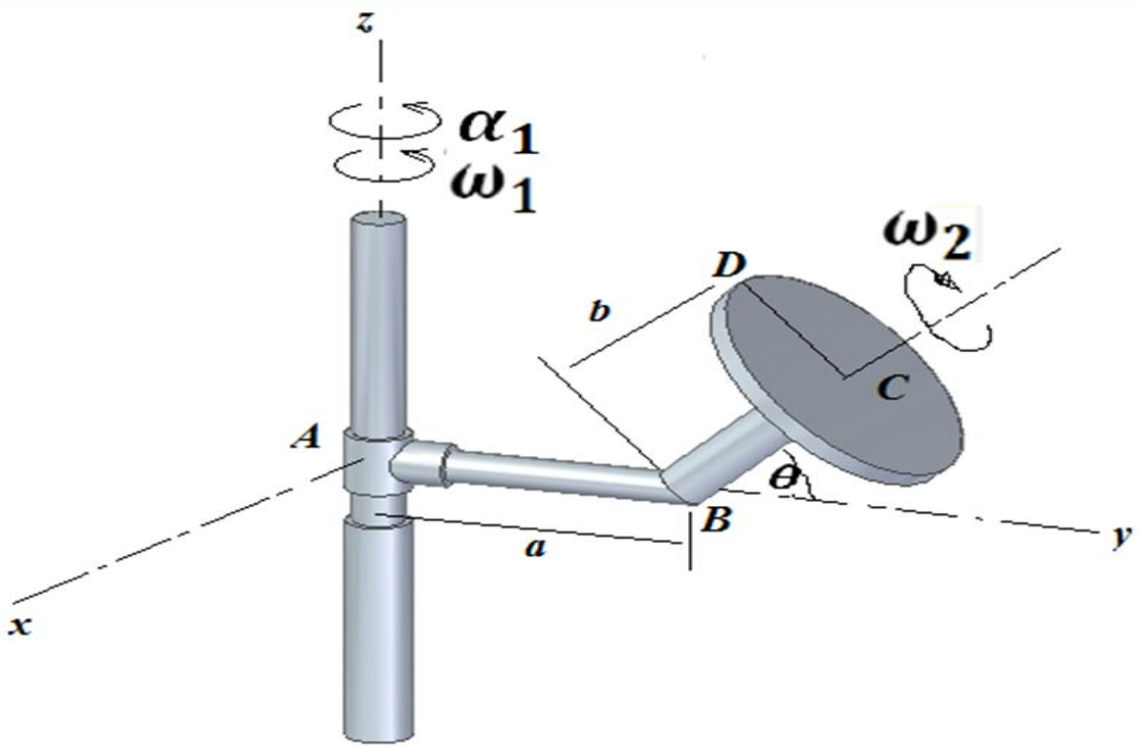
Knowing that $\theta = 30^o$, determine the

- angular velocity of the disc relative to the ground.
- angular acceleration of the disc relative to the ground.



Example 1-3 (Solution)

The angular velocity with reference to the fixed frame point B is simply the vector addition of angular velocities of AB about the vertical axis and disc about point C. The angular velocity of the disc with respect to point C is resolved into vertical and horizontal components. Notice that both the vertical and horizontal components of ω_2 are negative.



$$\omega = \omega_1 - \omega_2(\cos\ \theta\ \mathrm{j} + \sin\ \theta\ \mathrm{k}) = (3k) - 5(\cos\ 30^\circ\ \mathrm{j} + \sin\ 30^\circ\ \mathrm{k})$$

$$\omega = 4.33j + 0.5k$$

(b) angular acceleration

$$\alpha_1 = (\dot{\omega}_1)_{x,y,z} + 0\ \mathrm{x}\ \omega_1 = 1 \qquad \alpha_1 = 1\ \mathrm{k\ rad/s}$$

$$\alpha_2 = (\dot{\omega}_2)_{x,y,z} + \omega_1\ \mathrm{x}\ \omega_2 = (0) + (3k)\ \mathrm{x}\ [-5(\cos\ 30^\circ\ j + \sin\ 30^\circ\ k)] \qquad \alpha_2 = 12.99\ i$$

$$\alpha = \alpha_1 + \alpha_2 \qquad \alpha = 12.99\ i + k\ (\mathrm{rad/s}^2)$$

General Motion

Position Vector

$$r_B = r_A + r_{B/A}$$

Velocity

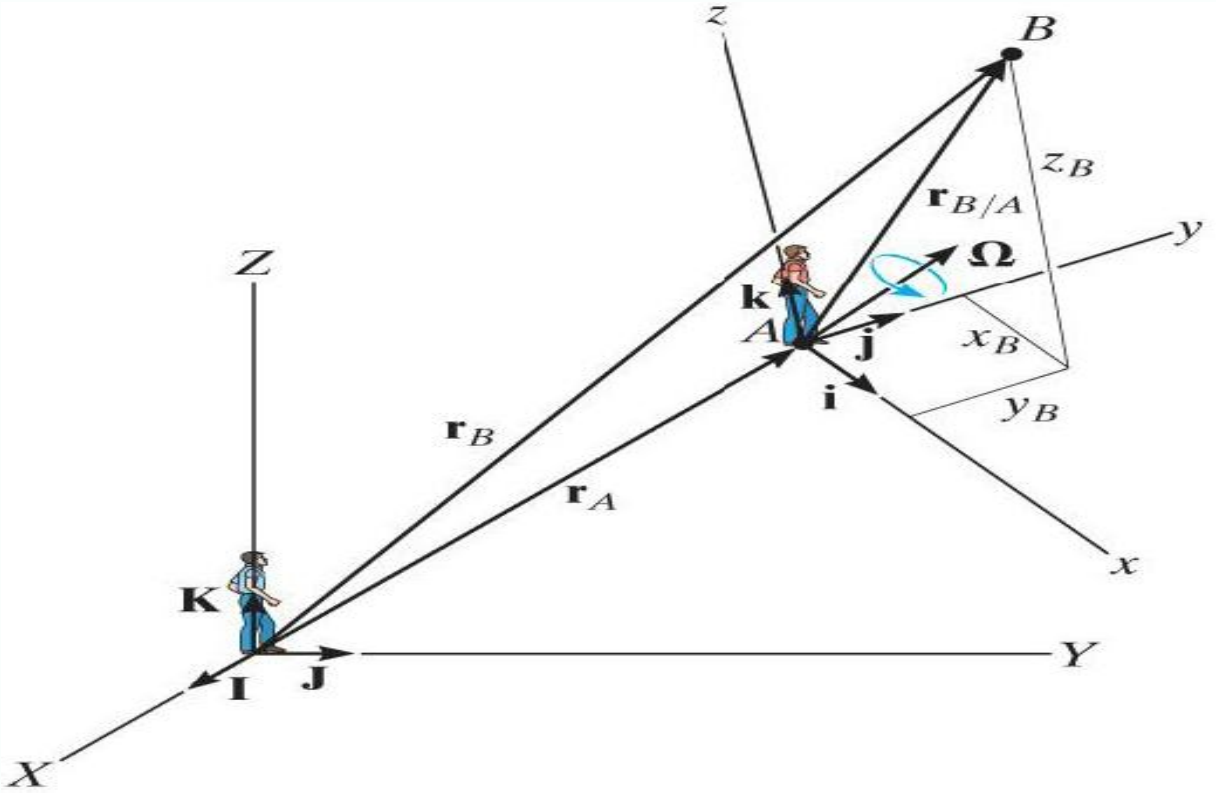
$$\frac{d}{dt}(r_B) = \frac{d}{dt}(r_A) + \frac{d}{dt}(r_{B/A})$$

$$v_B = v_A + (v_{B/A})_{xyz} + \Omega \times r_{B/A}$$

Acceleration

$$\frac{d}{dt}(v_B) = \frac{d}{dt}(v_A) + \frac{d}{dt}(v_{B/A})_{xyz} + \frac{d}{dt}(\Omega \times r_{B/A})$$

$$a_B = a_A + (a_{B/A})_{xyz} + \dot{\Omega} \times r_{B/A} + 2\Omega \times (v_{B/A})_{xyz} + \Omega \times (\Omega \times r_{B/A})$$



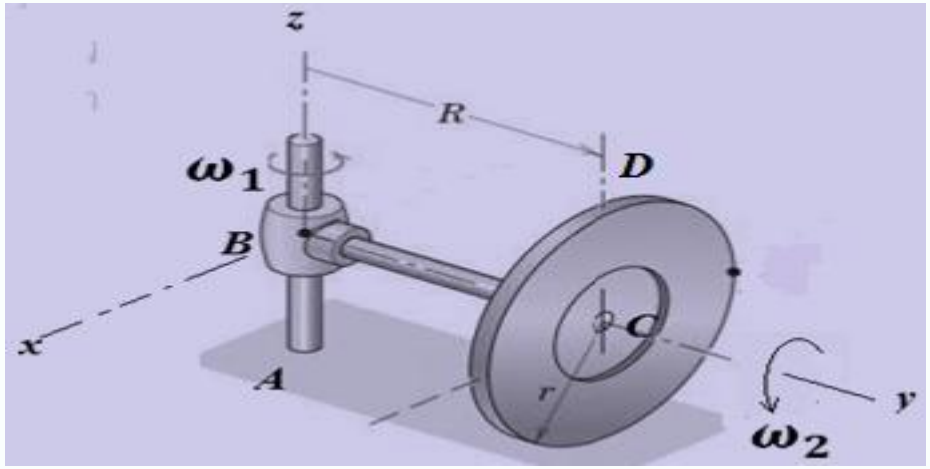
For 2D motions, the above equation reduces to

$$a_B = a_A + (a_{B/A})_{xyz} + \dot{\Omega} \times r_{B/A} + 2\Omega \times (v_{B/A})_{xyz} - \Omega^2 r_{B/A}$$

Example 1-4

Determine the velocity and acceleration of point D at the instant shown

R=1.5 m; r=0.5 m



Solution

From Example 1-2, absolute angular velocity and acceleration of the disc are

$$\omega = j + 4k \ (\text{rad/s})$$

$$\alpha = -4i + 2j \ (\text{rad/s}^2)$$

$$v_B = 0 \ \text{m/s}$$

$$v_C = v_B + v_{C/B} + \omega \times r_{C/B}$$

$$= 0 + 0 + (j + 4k) \times 1.5j = -6i \ \text{m/s}$$

The velocity of point D becomes

$$v_D = v_C + (v_{D/C})_{xyz} + \omega \times r_{D/C}$$

$$= -6i + 0 + (j + 4k) \times (0.5k) = -6i + 0.5i = -5.5i \ \text{m/s}$$

$$v_D = -5.5i \ (\text{m/s})$$

Acceleration analysis

$$a_B = 0 \ (\text{m/s}^2)$$

$$a_C = a_B + (a_{C/B})_{xyz} + \alpha \times r_{C/B} + 2\omega \times (v_{C/B})_{xyz} + \omega \times (\omega \times r_{C/B})$$

$$a_C = 0 + 0 + (-4i + 2j) \times 1.5j + 0 + (j + 4k) \times ((j + 4k) \times (1.5j))$$

$$a_C = (-4i + 2j) \times 1.5j + (j + 4k) \times (-6i)$$

$$a_C = -6k + 6k - 24j = -24j \ \text{m/s}^2$$

The acceleration of point D become

$$a_D = a_C + (a_{D/C})_{xyz} + \alpha \times r_{D/C} + 2\omega \times (v_{D/C})_{xyz} + \omega \times (\omega \times r_{D/C})$$

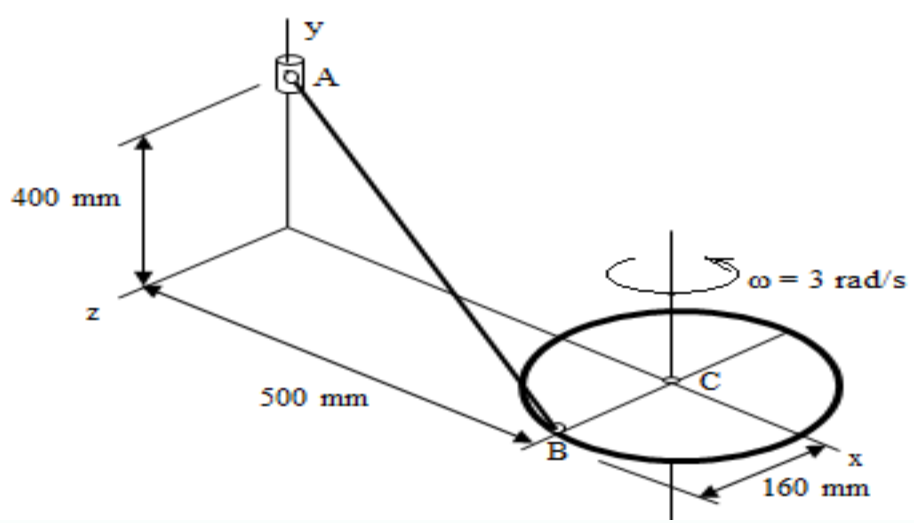
$$a_D = -24j + 0 + (-4i + 2j) \times (0.5k) + 0$$

$$+ (j + 4k) \times [(j + 4k) \times (0.5k)]$$

$$a_D = i - 20j - 0.5k \ (\text{m/s}^2)$$

Example 1-5

Rod AB is connected by ball-and-socket joints to collar A and to the 160-mm-radius disc C. Collar A slides on a vertical rod parallel to the y axis. If the disc rotates anticlockwise at a constant rate ω = 3 rad/s in the vertical plane, determine the (a) angular velocity of AB and linear velocity of collar A, and (b) angular acceleration of AB and linear acceleration of collar A for the position shown.



Solution

$v_C = 0$ m/s

$v_{B/A} = 0$

$v_B = v_C + \omega_{B/C} \times r_{B/C} = 0 + (3j \times 0.16k)$

$v_B = 0.48\,i$ m/s

Denoting the angular velocity of A relative to B by ω_{BA}, we have

Example 1-5, Conti;

$$v_A = v_B + \omega_{BA} \times r_{A/B} = 0.48\,i + \begin{vmatrix} i & j & k \\ \omega_x & \omega_y & \omega_z \\ -0.5 & 0.4 & -0.16 \end{vmatrix}$$

$$v_A = (0.48 - 0.16\,\omega_y - 0.48\,\omega_z)i + (0.16\,\omega_x - 0.50\,\omega_z)j + (0.40\,\omega_x + 0.50\,\omega_y)k \qquad (a)$$

Note that point A is attached to the collar and can only move in a direction parallel to the y axis. As a result, we have

$$v_A = V_A\,j \qquad (b)$$

Equating the coefficients of the unit vectors of equations (a) and (b), we have

$$0.16\,\omega_y + 0.48\,\omega_z = 0.48 \qquad (1)$$

$$0.16\,\omega_x - 0.50\,\omega_z = V_A \qquad (2)$$

$$0.40\,\omega_x + 0.50\,\omega_y = 0 \qquad (3)$$

Example 1-5, Conti;

The above three equations cannot be solved completely for the four unknowns. An additional equation is obtained by considering the constraint at point A. The connection at A allows rotation of BA about the vertical rod parallel to the y axis and about axis perpendicular to the plane containing the vertical rod and AB. It prevents rotation of AB about the axis AE illustrated below, which is perpendicular to vertical axis CD and lies in the plane of the vertical rod and AB. That is, the projection of angular velocity ω on $r_{E/A}$ must be zero, which yields

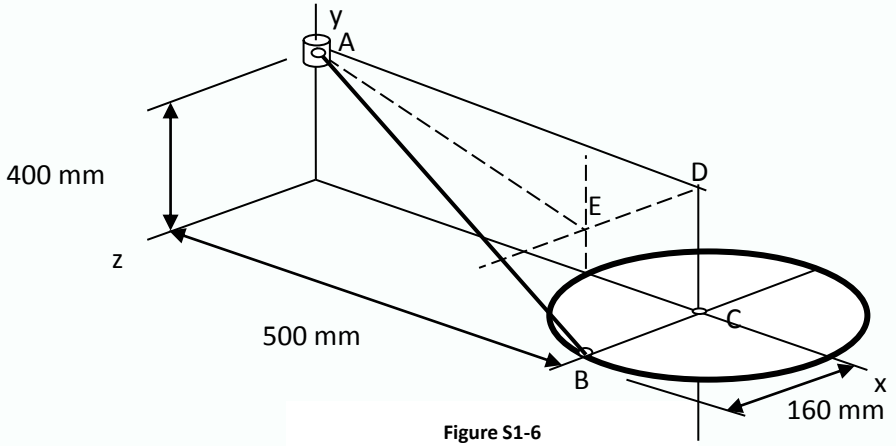


Figure S1-6

$$(\omega_x i + \omega_y j + \omega_z k) \bullet (0.5i + 0.16\,k) = 0$$

$$0.5\omega_x + 0.16\,\omega_z = 0 \qquad (4)$$

Solving equations (1) to (4) simultaneously, we have

$$\omega = (-0.348\,i + 0.279\,j + 1.089\,k)\ rad/s$$

$$v_A = -0.6\,j\ \text{m/s}$$

Example 1-5, Conti;

Note that $v_{A/B} = v_A - v_B = -0.6\,j - 0.48i$

The acceleration o point B is $a_B = a_C + \alpha \times r_{B/C} + \omega \times v_{B/C} = 0 + 0 + 3\,j \times 0.48i$

$$a_B = 1.44\,k \text{ m/s}^2$$

Denoting the angular acceleration of BA by α, we have

$$a_A = a_B + \alpha_{BA} \times r_{A/B} + \omega_{BA} \times v_{A/B} = 1.44\,k + \begin{vmatrix} i & j & k \\ \alpha_x & \alpha_y & \alpha_z \\ -0.5 & 0.4 & -0.16 \end{vmatrix} + \begin{vmatrix} i & j & k \\ -0.348 & 0.279 & 1.089 \\ -0.48 & -0.6 & 0 \end{vmatrix}$$

$$a_A = (0.653 - 0.16\,\alpha_y - 0.48\,\alpha_z)\,i + (-0.522 + 0.16\,\alpha_x - 0.50\,\alpha_z)\,j + (1.783 + 0.40\,\alpha_x + 0.50\,\alpha_y)\,k \quad (c)$$

Similar to the velocity, the point A can only move in a direction parallel to the y axis. As a result, we have $a_A = Aj$ (d)

Equating the coefficients of the unit vectors of Equation (c) to that of Equation (d), we have

Example 1-5, Conti;

$$0.653 - 0.16\,\alpha_y - 0.48\,\alpha_z = 0 \tag{5}$$

$$-0.522 + 0.16\,\alpha_x - 0.50\,\alpha_z = A \tag{6}$$

$$1.783 + 0.40\,\alpha_x + 0.50\,\alpha_y = 0 \tag{7}$$

Similar to the velocity, the projection of the angular acceleration on $r_{E/A}$ must be zero, which yields

$$(\alpha_x i + \alpha_y j + \alpha_z k).(0.5i + 0.16\,k) = 0$$

$$0.5\alpha_x + 0.16\,\alpha_z = 0 \tag{8}$$

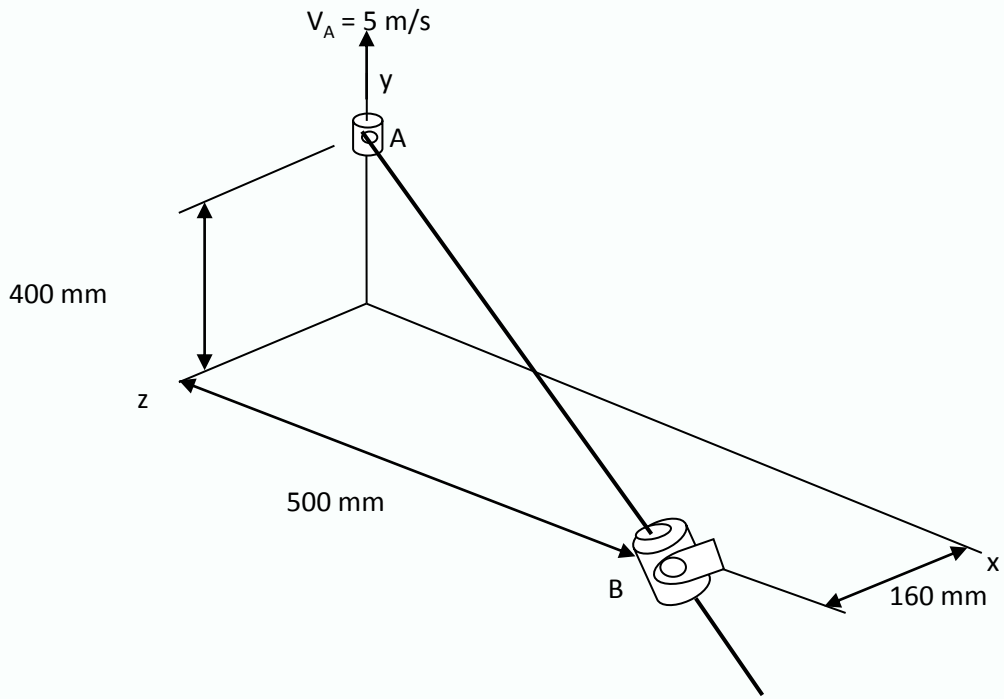
Solving Equations (5) to (8) simultaneously, we have

$$\alpha = (-0.752\ i - 2.965\ j + 2.349\ k)\ \text{rad/s}^2$$

$$a_A = -1.817\ j\ \text{m/s}^2$$

Example 1-6

Rod AB is connected by ball-and-socket joints to collar A and slides through sleeve B. Sleeve B is joined to the ground by a ball-and-socket joint . At the instant shown, the collar is sliding upward with a velocity of 5 m/s. Determine the (a) angular velocity of the rod AB, and (b) the velocity at which rod AB is sliding through sleeve B.



Example 1-6 (Solution)

$$v_A = v_B + (v_{A/B})_{xyz} + \omega \times r_{A/B} = 0 + (v_{A/B})_{xyz} + (v_{A/B})_{xyz} + \omega \times r_{A/B} = 5\,j \qquad (1)$$

Where $(V_{A/B})_{xyz}$ is the sliding velocity of rod AB through sleeve B, and ω is the angular velocity of AB. Let V be the magnitude of sliding velocity. then

$$(v_{A/B})_{xyz} = V\left\{\frac{-0.5i + 0.4j - 0.16k}{\sqrt{((-0.5)^2 + 0.4^2 + (-0.16)^2)}}\right\}$$

Let the angular velocity of AB be ω

$$\omega = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$$

Then, equation becomes

$$v_A = V\left\{\frac{-0.5i + 0.4j - 0.16k}{\sqrt{((-0.5)^2 + 0.4^2 + (-0.16)^2)}}\right\} + \begin{bmatrix} i & j & k \\ \omega_x & \omega_y & \omega_z \\ -0.5 & 0.4 & -0.16 \end{bmatrix} = 5j$$

Example 1-6 (Solution) Cont;

$$-0.16\,\omega_y - 0.4\,\omega_z - 0.7576\,V = 0 \qquad (1)$$

$$0.16\,\omega_x - 0.5\,\omega_z + 0.6061\,V = 5 \qquad (2)$$

$$0.4\,\omega_x + 0.5\,\omega_y - 0.2424\,V = 0 \qquad (3)$$

Similar to Example 1-5, the projection of angular velocity ω about the non-rotating axis must be zero, which yields

$$\omega \cdot(0.5\mathrm{i} + 0.16\mathrm{k}) = (\omega_x \mathrm{i} + \omega_y \mathrm{j} + \omega_z \mathrm{k}).(0.5i + 0.16\,k) = 0$$

$$0.5\,\omega_x + 0.16\,\omega_z = 0 \qquad (4)$$

Equations 1 to 4 may be grouped to form matrix given by

$$\begin{bmatrix} 0 & -0.16 & -0.4 & -0.7576 \\ 0.16 & 0 & -0.5 & 0.6061 \\ 0.4 & 0.5 & 0 & -0.2424 \\ 0.5 & 0 & 0.16 & 0 \end{bmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \\ V \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 0 \\ 0 \end{pmatrix}$$

Solving, we have

$$\begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \\ V \end{pmatrix} = \begin{pmatrix} 10.1150 \\ -18.0289 \\ -31.6095 \\ -20.4968 \end{pmatrix}$$

Solve for sliding and rotation accelerations of rod.

ASSIGNMENT 1

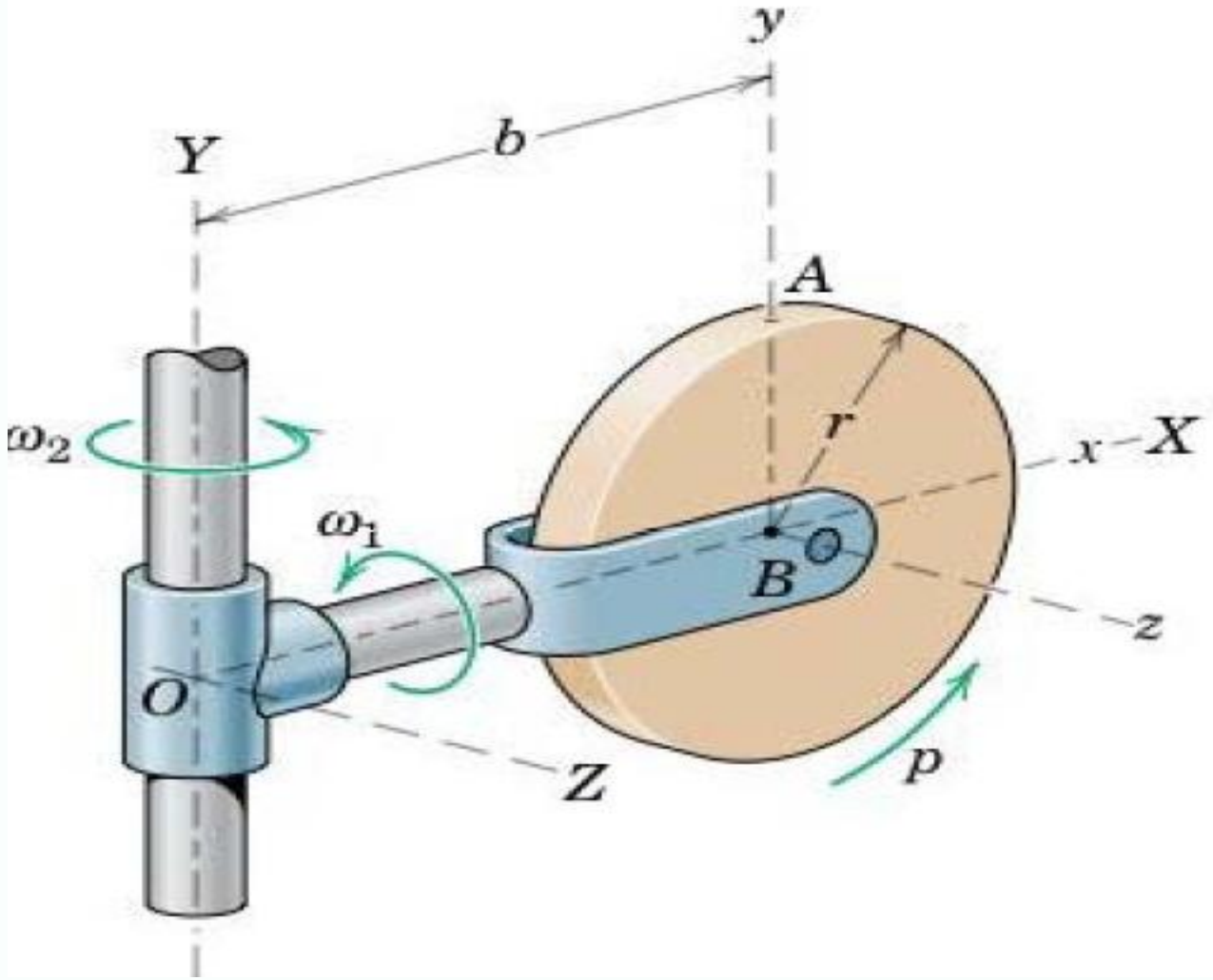
1.Determine the linear velocity and acceleration of point A on the disc shown Given:

$$\omega_1 = \frac{\pi}{10} rad/s$$

$$\omega_2 = \frac{\pi}{5} rad/s$$

$$p = \pi\ rad/s$$

r = 2 m and b = 4.5 m

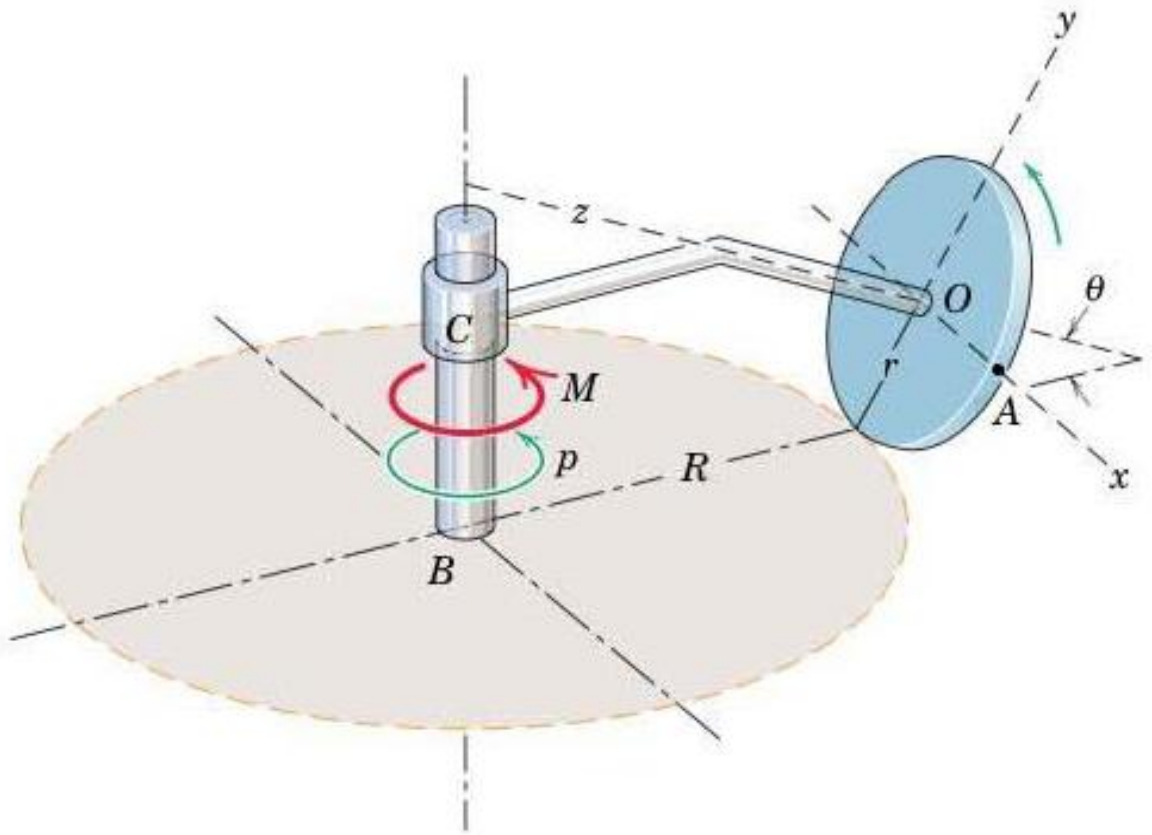


ASSIGNMENT 2

2. Determine the linear velocity and acceleration of point A on the disc shown
Given

$$R = 100\ mm, r = 50\ mm$$

$$\theta = 60^o$$
$$p = \pi\ rad/s$$



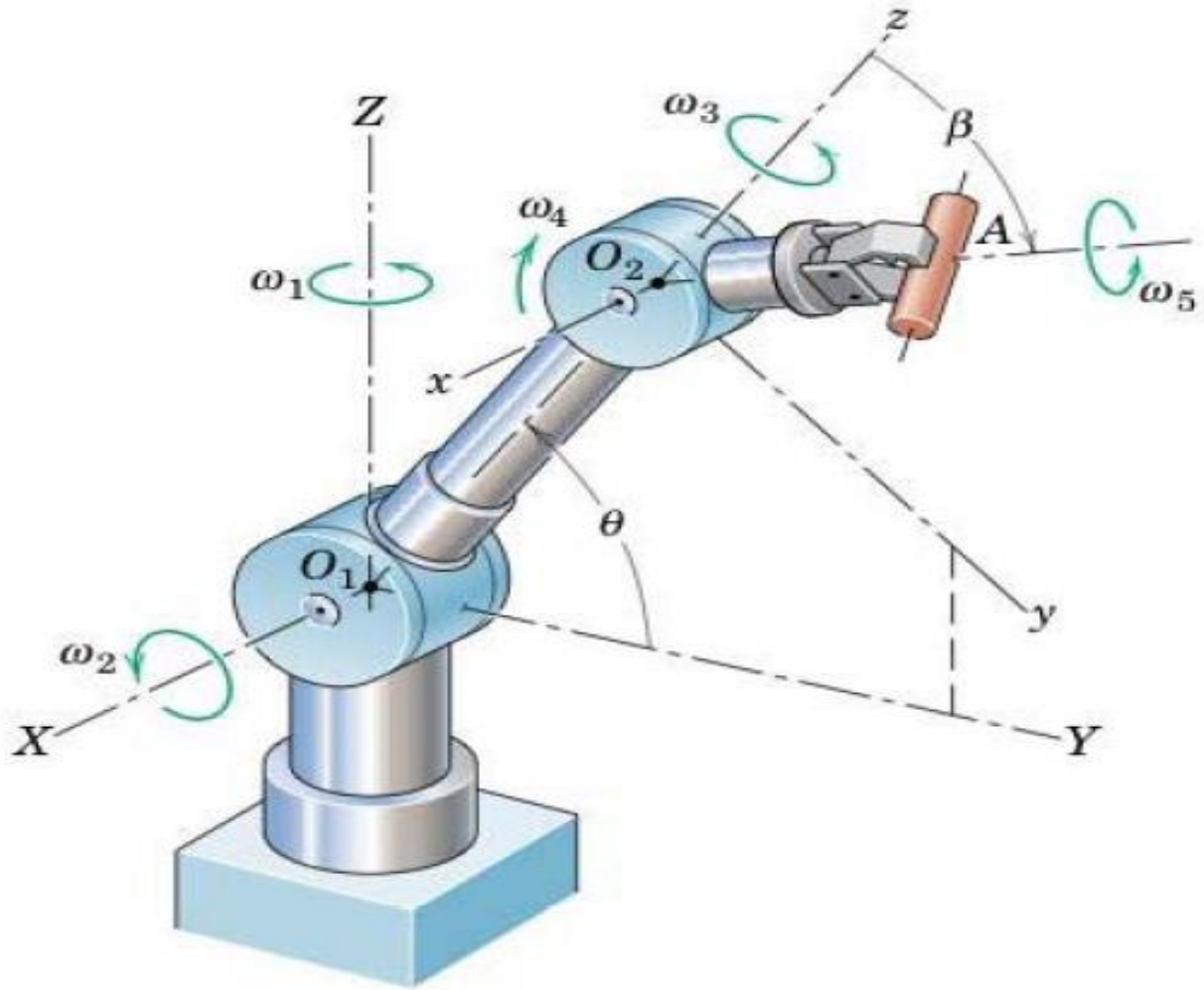
ASSIGNMENT 3

The robot shown has:

$$\omega_1 = \omega_5 = 2\,rad/s,$$
$$\omega_2 = 3.5\,rad/s,$$
$$\omega_3 = \omega_4 = \pi\,rad/s$$
$$O_1O_2 = 1.2\,m, O_2A = 0.6\,m$$

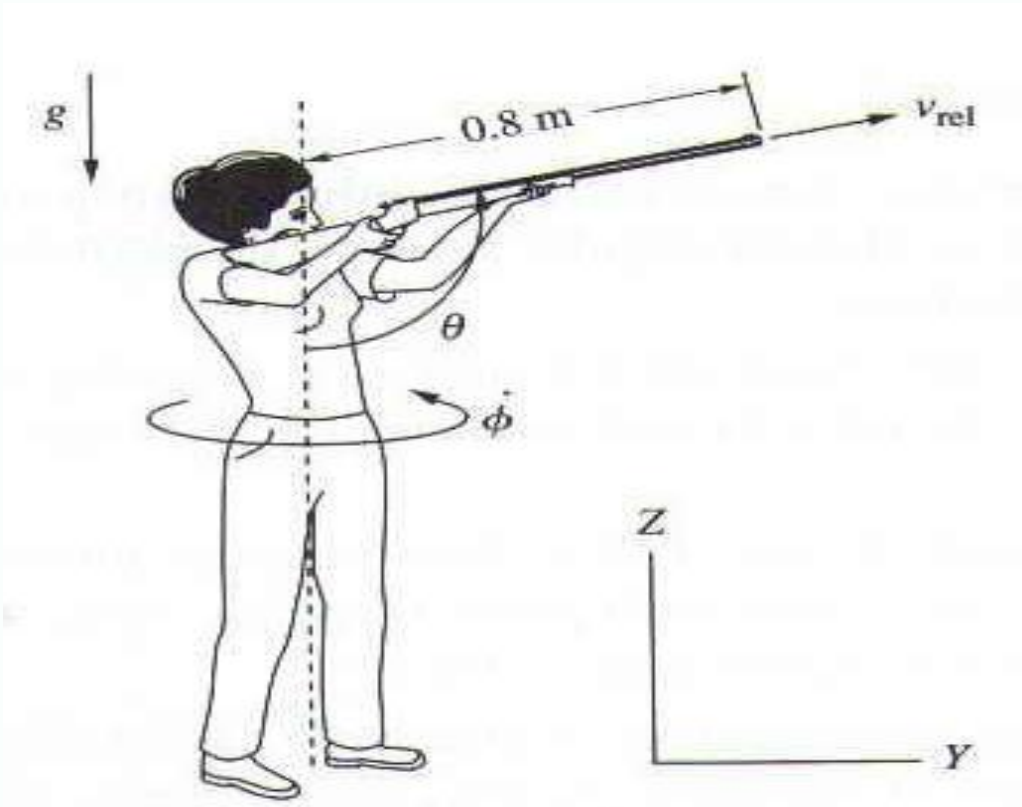
Determine the velocity of the center A of the jaws for the instant when $\theta = 60^o$.

The angle β lies in the y-z plane and is constant at 45°.



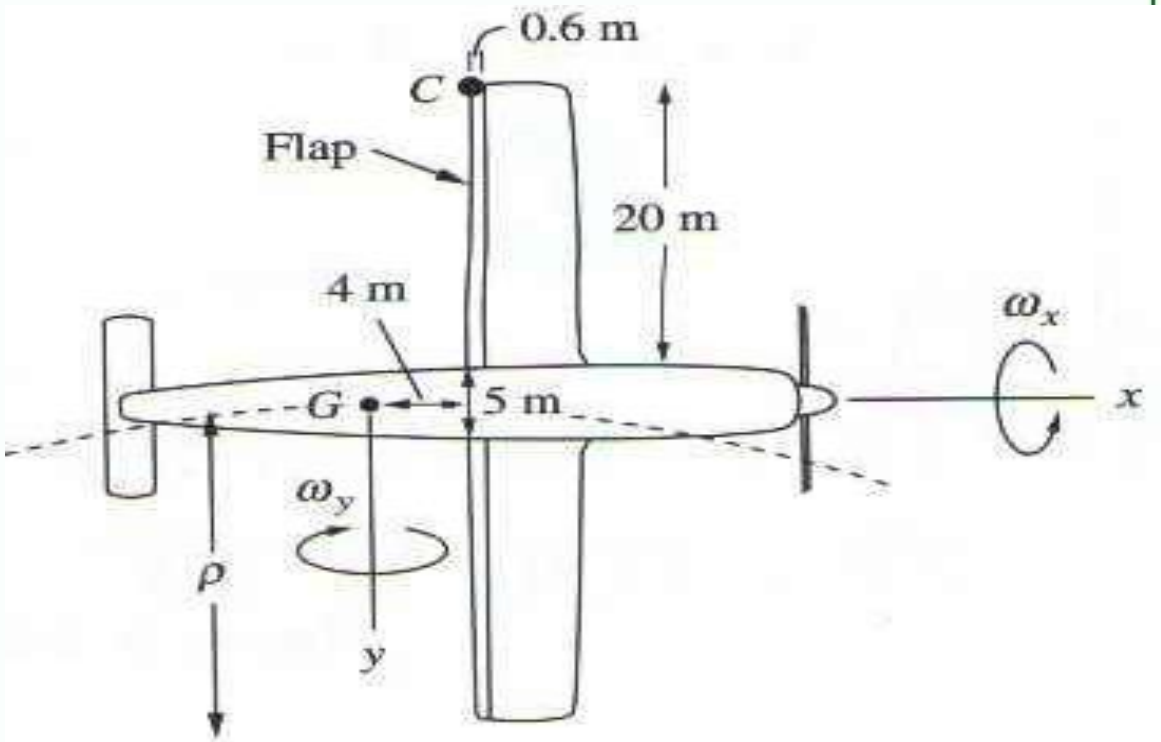
ASSIGNMENT 4

A hunter shown in the Figure is aiming at a moving target with her rifle. The hunter is moving the rifle upwards with a constant angular velocity of $\dot{\theta} = 10°/s$ and pulls the trigger when θ= 105°. The bullet, which weighs 1/8 kg, travels in the barrel with a constant speed of 100 km/h relative to the rifle. Find the velocity and acceleration of the bullet.



ASSIGNMENT 5

An airplane, shown in the Figure, is executing a circular turn with a radius of 4000 m, while flying at a speed of 600 km/h. The airplane has a constant pitch rate of $\omega_y = 0.04\, rad/s$ and a constant roll rate of $\omega_x = 0.01\, rad/s$. The trailing edge flaps are being extended to give the airplane more lift at the constant rate of 0.05 m/s. Find the velocity and acceleration of point C on the flap.



ASSIGNMENT 6

The collars *A* and *B* are attached by a rod of length 30 cm, as shown in the Figure. The joint at *A* is a ball and socket joint and at *B* a pin joint. Collar A moves in the Z direction, while the guide bar for collar *B* is on the X *Y* plane. At the point shown, collar A is at a height of 24 cm and collar *B* is moving with a speed of 10 cm/s. Find the angular velocity of the rod.

